In [21]:
```
#creates family dataframe
import pandas as pd
testdata = pd.read_csv("C:/Users/keert/OneDrive/Desktop/keerti/module6/IUCNHierarch
family = testdata[['family']].copy()
family = family ["family"].str.strip()
family = family.map(str.lower)
duplicate_family = family[family.duplicated()]
family = family.drop_duplicates()
family = family.rename({'family':'fName'})
```

In [22]:
```
#creates genus dataframe
genus = testdata[['genus', 'family']].copy()
genus['genus'] = genus["genus"].str.strip()
genus['family'] = genus["family"].str.strip()
genus.columns = genus.columns.map(str.lower)
duplicate_genus = genus[genus.duplicated()]
genus = genus.drop_duplicates()
genus = genus.rename(columns={'genus':'gName', 'family':'fName'})
```

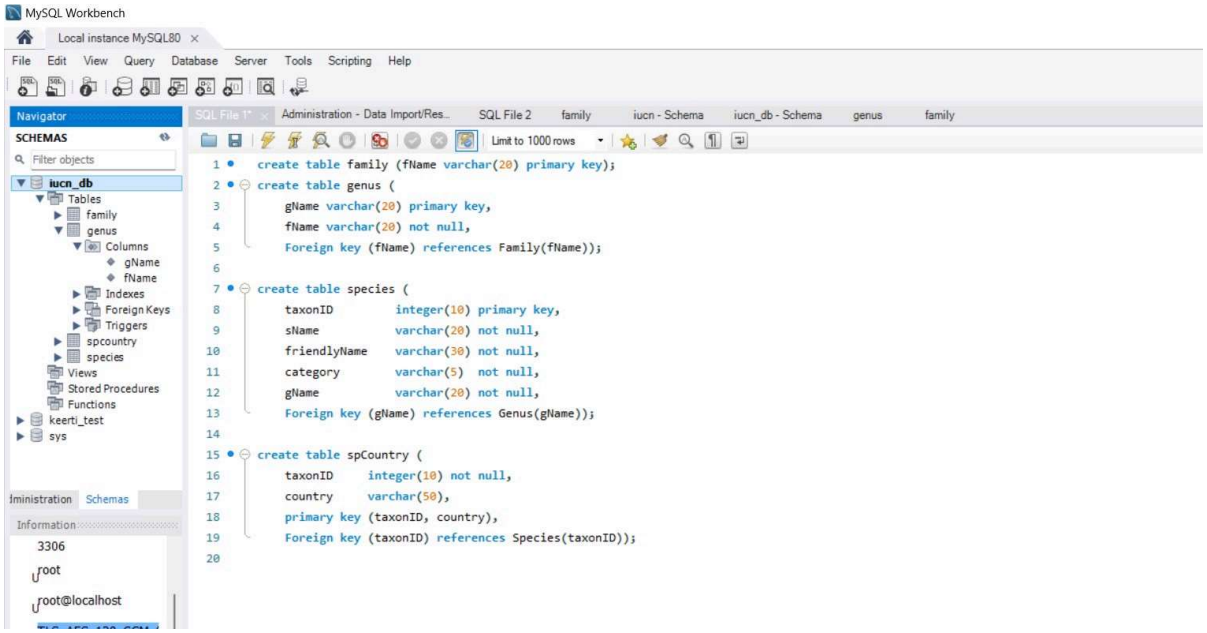
In [6]:
```
#creates species dataframe
species = testdata[['taxonid', 'species', 'friendly_name', 'category', 'genus']].co
species.columns = species.columns.map(str.lower)
duplicate_species = species[species.duplicated()]
species = species.drop_duplicates()
species = species.rename(columns={'taxonid':'taxonID', 'species':'sName', 'friendly
```

In [52]:
```
#creates spCountry dataframe
#SpCountry (taxonID, country)
testdata2 = pd.read_csv("C:/Users/keert/OneDrive/Desktop/keerti/module6/IUCNSpecies
spCountry = testdata2[['taxonid', 'countryoccurrencelookup_value']].copy()
spCountry.columns = spCountry.columns.map(str.lower)
duplicate_spCountry = spCountry[spCountry.duplicated()]
spCountry = spCountry.drop_duplicates()
spCountry = spCountry.rename(columns={'taxonid':'taxonID', 'countryoccurrencelookup
```

In [53]:
```
#saving the dataframes as csv files
family.to_csv('family.csv', index=False)
genus.to_csv('genus.csv', index=False)
species.to_csv('species.csv', index=False)
spCountry.to_csv('spCountry.csv', index=False)
```

In [24]:
```
from IPython.display import Image
Image(filename="C:\\Users\\keert\\OneDrive\\Desktop\\keerti\\module6\\mysqlTableDat
```

Out[24]:



In [3]:

```
# import statements
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [7]:

```
#reads in threatenedSpeciesPerFamily from csv
threatenedSpecies = pd.read_csv('C:\\Users\\keert\\OneDrive\\Desktop\\keerti\\modul
```

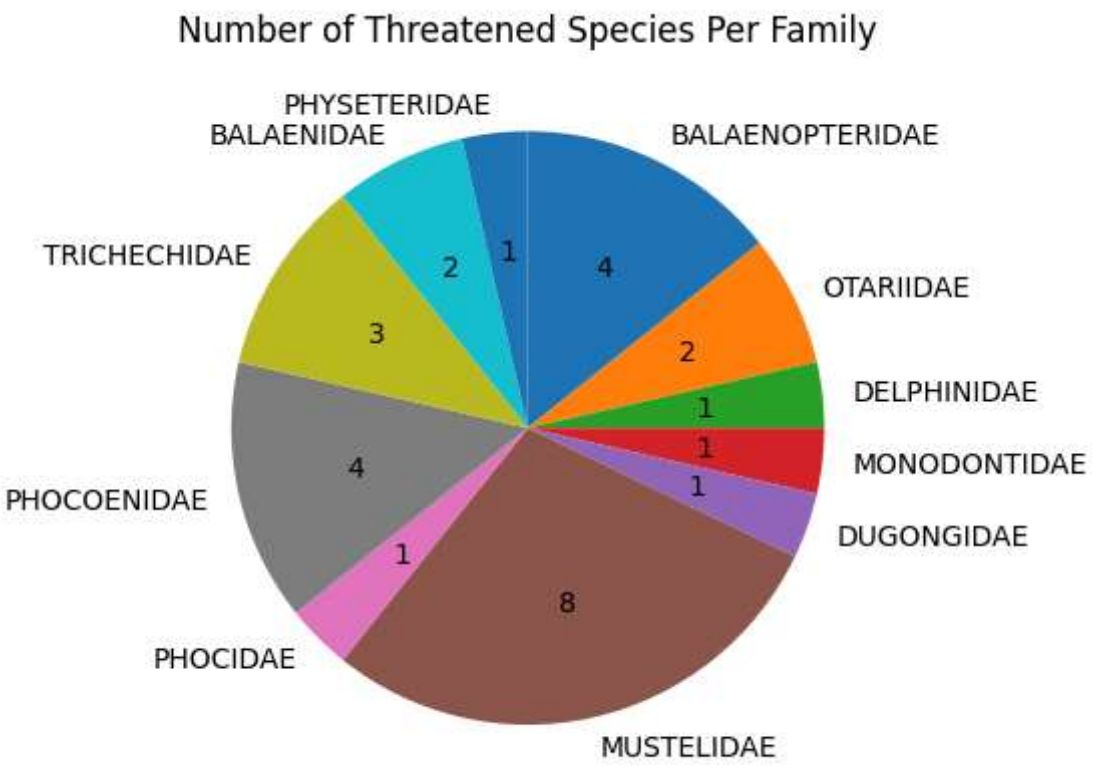
In [6]:

```
# get values from the DataFrame
fName = threatenedSpecies['fName']
numberOfThreatenedSpecies = threatenedSpecies['numberOfThreatenedSpecies']

# plot the pie chart
plt.pie(numberOfThreatenedSpecies, labels=fName, startangle=90, counterclock=False,

# add information to the plot
plt.title('Number of Threatened Species Per Family')

# show the plot
plt.show()
```


Number of Threatened Species Per Family
PHYSETERIDAE
BALAENIDAE
BALAENOPTERIDAE
TRICHECHIDAE
OTARIIDAE
DELPHINIDAE
MONODONTIDAE
PHOCOENIDAE
DUGONGIDAE
PHOCIDAE
MUSTELIDAE
1
2
4
3
2
1
1
1
4
1
8

In [ ]:

In [ ]:

In [ ]:

In [ ]: